SMOOTHING CALABI–YAU THREEFOLDS IN GORENSTEIN TORIC FANO FOURFOLDS

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Abstract. Let $\Delta \subset \mathbb{Z}^4$ be a reflexive polytope with unit edges and $X \subset \mathbb{P}_\Delta$ a generic anticanonical hypersurface in the Gorenstein toric Fano fourfold of the face fan of Δ: a Calabi–Yau threefold whose singular points are, for each non-smooth two-dimensional face F, the $\ell(F^\circ)$ points at which the germ of X is the affine Gorenstein toric threefold $C(F)$, the cone over F at height one, $\ell(F^\circ)$ being the lattice length of the edge of Δ° dual to F. Its predecessor (*Non-smoothable Calabi–Yau threefolds from reflexive polytopes*) treated the faces whose germs have no smoothing component; here every germ is locally smoothable and the question is global. We prove that the points over a face of type (S), a Minkowski sum of unit segments and standard triangles, are smoothed by an explicit deformation of the ambient toric pair, cut out by trinomials in Cox coordinates, whenever $\ell(F^\circ) \geq 2$, transplanting Petracci's homogeneous deformations of toric pairs to codimension-two faces. Hence X is smoothable when F is its unique non-smooth face; unconditionally when every non-smooth face is a unit parallelogram with $\ell(F^\circ) \geq 2$; and, for several type-(S) faces, whenever the semiuniversal deformation space of X is irreducible. The threshold is sharp: every generic unit-edge Batyrev hypersurface whose singular locus consists of a single ordinary double point admits no smoothing, and we exhibit one from a reflexive polytope with 7 vertices. At $\ell(F^\circ) = 1$ the germs over a face with one interior lattice point are the anticanonical cones over $\mathbb{P}^1 \times \mathbb{P}^1$, dP_7 and dP_6; we show their toric crepant resolutions are *never* primitive contractions and that X is not $\mathbb{Q}$-factorial there, so neither Friedman's criterion nor Gross's smoothing theorems decide them, and they remain open. Finally we compare the resulting per-face criterion with the census of Batyrev–Kreuzer, whose criterion constrains only the $\ell(F^\circ) = 1$ faces. Subject to the database-copy completeness hypothesis stated in §7, exactly 3,774 of their 30,241 smoothings, one in eight, are explained by the per-face criterion alone, and the remaining 26,467 involve a cross-face homology relation.

1. Introduction

Let $\Delta \subset \mathbb{Z}^4$ be a reflexive polytope all of whose edges have lattice length one, and let $X \subset \mathbb{P}_\Delta$ be a generic anticanonical hypersurface in the Gorenstein Fano toric fourfold associated to the face fan of Δ. By Batyrev [Bat94] X is a Calabi–Yau threefold with Gorenstein canonical singularities, and by [Wue26a, Prop. 3.1] its singular locus is exactly: for each two-dimensional face $F \preceq \Delta$ whose cone $C(F)$ is singular, $\ell(F^\circ)$ points at which the germ of X is analytically isomorphic to $C(F)$.

The deformation theory of the germs is completely understood combinatorially: by Altmann [Alt97] and Corti–Filip–Petracci [CFP22], $C(F)$ is *smoothable* if and only if F admits a Minkowski decomposition into unit segments and standard triangles (type (S) in the

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Altmann–Corti–Filip–Petracci trichotomy, as recalled in [Wue26a, Thm. 2.2]). The companion paper proved that a single face of the other two types (rigid or deformable-but-non-smoothable) forbids every global smoothing of X. This paper treats the remaining case: *every* singular two-dimensional face is of type (S), so every singular point of X admits a local smoothing, and the question is whether the local smoothings are realized by a global deformation. This is the classical local-to-global problem: for ordinary double points smoothability is governed by Friedman's relation among the exceptional curve classes [Fri86, RT09], and for the toric hypersurfaces at hand it was studied, in the all-conifold case, by Batyrev–Kreuzer [BK10], who found that of the 198,849 reflexive 4-polytopes whose two-dimensional faces are standard triangles and unit parallelograms (at least one parallelogram), only 30,241 could be shown smoothable by Namikawa's criterion. One dimension down, the analogous constructive problem is to smooth a Gorenstein toric Fano *threefold* from Minkowski data on the facets of a reflexive 3-polytope; it was recently solved by Corti–Hacking–Petracci [CHP24]. The obstruction side of the same programme is due to Petracci [Pet20], who produces non-smoothable Gorenstein toric Fano threefolds by a local-to-global criterion on a reflexive 3-polytope. Theorem 1.3 below plays the corresponding role here: it gives a uniform one-node obstruction and an explicit reflexive 4-polytope realizing it. In both of those works the Minkowski data live on the *facets* of the 3-polytope and the object smoothed or obstructed is the Gorenstein toric Fano threefold itself; here the data live on the codimension-two faces of a reflexive 4-polytope and the object smoothed is the anticanonical hypersurface in the toric fourfold.

Our first result is constructive and identifies a clean sufficient condition. Write $\ell(F^\circ)$ for the lattice length of the edge of Δ° dual to a two-dimensional face F.

Theorem 1.1 (= Theorem 4.1)**.** *Let Δ be reflexive with unit edges and let F be a two-dimensional face of type* (S)*, with a Minkowski decomposition $F = Q_0 + \cdots + Q_r$ into unit segments and standard triangles, and suppose $\ell(F^\circ) \geq 2$. Then there is an explicit r-parameter deformation of the pair $(\mathbb{P}_\Delta, \partial\mathbb{P}_\Delta)$, given by trinomial equations in Cox coordinates, whose induced deformation of the generic anticanonical X has generic fibre smooth in a neighbourhood of the $\ell(F^\circ)$ points of type $C(F)$. In particular, if F is the* unique *non-smooth two-dimensional face of Δ, then X is smoothable.*

The idea is simple to state. The deformation is Petracci's homogeneous deformation of the polarized toric pair, applied not to a facet but to the codimension-two face F: the deformation datum lives on the cone over the pair, with weight an *interior* lattice point v^* of the dual edge F°. At the two endpoints of F° the relevant slice of the cone grows to a full facet of Δ, and the datum would require a Minkowski decomposition of that three-dimensional facet; the interior points of F° are exactly what makes a face-local datum possible. This is the source of the threshold $\ell(F^\circ) \geq 2$, and §5 shows the threshold is sharp.

Theorem 1.1 belongs to a known genre. A Calabi–Yau hypersurface in a toric variety has *polynomial* deformations, obtained by moving X inside a fixed $\mathbb{P}_\Delta$, and these never change the singularity structure here (Remark 5.2); the *non-polynomial* deformations, induced instead from deformations of the ambient toric variety, were constructed by Mavlyutov [Mav04]. The construction we use is Petracci's [Pet21], which extends this Altmann–Mavlyutov theory of homogeneous deformations [Alt95, Mav04] from affine toric varieties to toric pairs, and Theorem 1.1 is of exactly this non-polynomial kind. What is new is not the passage from ambient to hypersurface but the *source* of the ambient deformation: a datum supported on a single codimension-two face, whose existence is governed by the lattice length of the dual edge and whose effect on the germ is read off from the Minkowski decomposition of that face.

This is what makes the criterion face-local, and hence testable across the Kreuzer–Skarke list.

For several type-(S) faces the construction smooths the points over each face individually; the face-wise smoothings assemble unconditionally in the nodal case, and in general under a hypothesis on the deformation space. (The fibres of the resulting family are no longer toric, so the construction cannot simply be iterated; see Remark 4.6.)

Theorem 1.2 (= Theorem 4.5)**.** *Let Δ be reflexive with unit edges and suppose that every non-smooth two-dimensional face $F_1, \dots, F_s$ of Δ is of type* (S) *(a Minkowski sum of unit segments and standard triangles), with $\ell(F_i^\circ) \geq 2$. Then:*

(1) *for every singular point p of X there is an explicit deformation of X whose generic fibre is smooth in a neighbourhood of p; consequently the locus $D_p \subset (B, 0)$ of points of the semiuniversal deformation of X whose fibres remain singular near p is a proper closed analytic subgerm;*
(2) *if the base germ $(B, 0)$ of the semiuniversal deformation of X is irreducible (for instance smooth), then X is smoothable;*
(3) *if every F_i is a unit parallelogram, then X is smoothable unconditionally.*

Parts (ii) and (iii) are proved by different mechanisms. Part (ii) is soft. By (i) each singular point is removed by *some* deformation, so the locus in the semiuniversal base over which that point stays singular is a proper closed analytic subgerm; the singular points are finite in number, so there are finitely many such loci. A finite union of proper closed analytic subgerms cannot exhaust an irreducible germ, and an arc through the origin missing all of them is a smoothing. Irreducibility is thus precisely what converts "removable one point at a time" into "removable simultaneously", and it is a hypothesis we cannot presently discharge: by Gross [Gro97] the deformation space of a Calabi–Yau threefold with canonical singularities need not be smooth, and we do not know whether the base is irreducible for the hypersurfaces at hand (Question 8.2). Part (iii) avoids the hypothesis altogether by working in homology instead of in the deformation space. For nodes, the $\ell(F_i^\circ)$ exceptional curves over one face all carry the *same* class, so Friedman's relation can be satisfied one face at a time: one chooses non-zero integer coefficients summing to zero, which is possible exactly when $\ell(F_i^\circ) \geq 2$, and the per-face relations simply add. This is the same threshold as Theorem 1.1, arrived at from homology rather than from the existence of a deformation datum, and §7 takes up the coincidence.

The type-(S) hypothesis enters only at the last step, to make the local general fibre smooth. Applied to an *arbitrary* Minkowski decomposition the same face family therefore realizes globally the cascade of [Wue26a, Lemma 2.4], replacing each of the $\ell(F^\circ)$ points over F by one point of type $C(Q_i)$ for every two-dimensional non-smooth summand Q_i (Proposition 4.7). For the deformable-but-non-smoothable pentagon of [Wue26a] this yields an explicit trinomial deformation between two *non-smoothable* Calabi–Yau threefolds, each of three deformable-but-non-smoothable germs being traded for a rigid $\frac{1}{3}(1,1,1)$ point. This is an obstructed analogue of a conifold transition, in which the deformation genuinely moves but no fibre is ever smooth (Remark 4.8).

The hypothesis $\ell(F^\circ) \geq 2$ is not an artifact of the method.

Theorem 1.3 (= Theorem 5.1)**.** *Let Δ be a reflexive 4-polytope with unit edges. If the singular locus of its generic anticanonical hypersurface $X \subset \mathbb{P}_\Delta$ consists of a single ordinary double point, then X admits no smoothing.*

In particular, let

$$\Delta_N = \operatorname{conv}\big(\square \times \{(1,0)\} \ \cup\ \{e_4,\ (0,1,-1,0),\ (-1,-2,1,-1)\}\big) \subset \mathbb{Z}^4,$$

where $\square$ *is the unit square embedded with vertices* $(0,0),(0,-1),(1,-1),(1,0)$*. Then* Δ_N *is reflexive with* 7 *vertices and unit edges; its unique non-smooth two-dimensional face is* $\square \times \{(1,0)\}$*, of type* (S)*, with dual edge of lattice length* 1*; the generic anticanonical* X_N *is a Calabi–Yau threefold with exactly one ordinary double point; and* X_N *admits no smoothing. The maximal projective crepant partial (MPCP) resolution* $\widehat{X}_N$ *has* $(h^{1,1},h^{2,1}) = (3,103)$ *and Euler number* -200*.*

Thus at $\ell(F^\circ) = 1$ the local-to-global obstruction is real even though the local model is the simplest smoothable singularity there is. The obstruction is Friedman's: the maximal projective crepant partial resolution of X_N is a *projective small* resolution, so the exceptional curve class cannot vanish, while a one-node Calabi–Yau threefold is smoothable only if it does. By contrast, for germs whose crepant resolution is divisorial, Friedman's obstruction has no analogue, and no other is known:

Proposition 1.4 (= Proposition 6.1)**.** *Let* Δ *be reflexive with unit edges whose unique non-smooth two-dimensional face* F *has exactly one interior lattice point, is of type* (S)*, and has* $\ell(F^\circ) = 1$*. Then the singular locus of* X *is one point, the anticanonical cone over* $S \in \{\mathbb{P}^1 \times \mathbb{P}^1,\ \mathrm{dP}_7,\ \mathrm{dP}_6\}$*, and an MPCP resolution* $\widehat{X} \to X$ *can be chosen whose exceptional locus is the smooth irreducible surface* S*, contracted to the point. However, the contraction has relative Picard number* $\rho(\widehat{X}/X) = \rho(S) \in \{2,3,4\}$*, so it factors in the projective category and is* never *a primitive contraction, and* X *is not* $\mathbb{Q}$*-factorial at the point. In particular neither Gross's theorem on primitive type II contractions* [Gro97, Thm. 5.8] *nor his smoothing theorem for* $\mathbb{Q}$*-factorial contractions* [Gro97, Thm. 4.3] *applies, and the smoothability of* X *is open (Question 8.1).*

The statements assemble into a germ-dependent picture of the local-to-global problem (Table 1): for type-(S) faces the dual-edge length $\ell(F^\circ) \geq 2$ smooths the points over the face, and the face-wise smoothings assemble in the nodal case and, in general, whenever the deformation space of X is irreducible (Theorem 1.2); at $\ell(F^\circ) = 1$ the picture is governed by the crepant resolution over the point. For the node the resolution is small and Friedman's relation obstructs (Theorem 1.3). For the germs over a face with a single interior lattice point ($i(F) = 1$) the resolution is divisorial, but, in contrast to the primitive type II contractions analyzed by Gross, Proposition 1.4 shows the toric contraction always has relative Picard number ≥ 2 and X is not $\mathbb{Q}$-factorial, so the two known mechanisms (Friedman's obstruction; Gross's smoothing theorems) both fall silent and the case is open. At the level of germs the alignment is still pleasant: Gross's theorem for primitive realizations fails exactly when the exceptional surface is $\mathbb{P}^2$ or $\mathbb{F}_1$, and these are precisely the two classes with one interior point that are *rigid* in the trichotomy (Theorem 2.1). The local trichotomy and the global smoothing problem see the same dichotomy.

Section 7 confronts this picture with the Batyrev–Kreuzer census. For the $\ell(F^\circ)$ nodes over a single square face the exceptional classes coincide, so Friedman's all-nonzero relation is satisfiable face-locally if and only if $\ell(F^\circ) \geq 2$; this reproduces the threshold of Theorem 1.1 from pure homology. A scan of the database copy described in §7, subject to the completeness hypothesis stated there, shows that this per-face condition accounts for exactly 3,774 of the 30,241 conifold polytopes that Batyrev–Kreuzer show to be smoothable, one in eight, so the great majority of their smoothings are *cross-face* rescues.

germ of the type-(S) face	$\ell(F^\circ) = 1$	$\ell(F^\circ) \geq 2$
unit square (node; crepant res. small)	**non-smoothable** (Thm 1.3)	smoothable (Thm 1.1)
$i(F) = 1$ (cone over $\mathbb{P}^1 \times \mathbb{P}^1, \mathrm{dP}_7, \mathrm{dP}_6$; divisorial)	open: never primitive torically (Prop. 1.4)	smoothable (Thm 1.1)
$i(F) \geq 2$ type (S) (divisorial, reducible exceptional locus)	open	smoothable (Thm 1.1)

TABLE 1. The local-to-global picture by germ and dual-edge length, for a single singular face. For several singular faces see Theorem 1.2: the points over each face are smoothed individually, the all-conifold case is unconditional, and the general case assembles when the deformation space is irreducible; mixed configurations involving $\ell = 1$ nodes are governed by cross-face Friedman relations (§7).

All computations are performed by the exact-arithmetic programs of the companion paper, extended by a routine enumerating the conifold polytopes of the Kreuzer–Skarke list; they are included as ancillary files and are public, with all data, at

https://github.com/bwuebben/calabi-yau-smoothability.

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2. PRELIMINARIES

We keep the notation of [Wue26a] and recall here everything that is used. A *standard triangle* is a lattice triangle equivalent to $\mathrm{conv}\{0, e_1, e_2\}$, and a *unit segment* is a lattice segment of lattice length one; a *unit parallelogram* is a lattice parallelogram equivalent to the unit square, and we use the two words interchangeably, its cone being the ordinary double point. For a lattice polygon Q with unit edges placed at height one, $C(Q)$ denotes the associated Gorenstein toric affine threefold, and $i(Q)$ the number of interior lattice points of Q. (Throughout, $C(\cdot)$ *with an argument* denotes this affine germ; exceptional curves are written γ, γ_i, γ_F, so that the two conventions in force here, Altmann's $C(\cdot)$ for the cone and Friedman's C_i for the exceptional curves [Fri86, BK10], do not collide.) The reduced miniversal base of $C(Q)$ has one smooth component per maximal Minkowski decomposition of Q [Alt97], the smoothing components corresponding to decompositions into unit segments and standard triangles [CFP22]; Q is of *type (S)* when such a decomposition exists. The four imported statements used below are the following; we give each a number, and refer to it by that number rather than by name.

Theorem 2.1 (the trichotomy; [Wue26a, Thm. 2.2], packaging [Alt97, CFP22])**.** *Let Q be a unit-edge lattice polygon that is not a standard triangle, so that $C(Q)$ is singular. Then Q is of exactly one of three types:* (S)*, admitting a Minkowski decomposition into unit segments and standard triangles, so that $C(Q)$ is smoothable;* (R) (rigid)*, admitting no non-trivial Minkowski decomposition, so that $C(Q)$ has no positive-dimensional deformations (the reduced miniversal base is a point; first-order deformations may exist); or* (D)*, admitting a non-trivial decomposition but none into unit segments and standard triangles, so that $C(Q)$ deforms but is not smoothable.*

Lemma 2.2 (local-to-global obstruction; [Wue26a, Lemma 4.1])**.** *A deformation of X over a reduced irreducible base induces, near each singular point, a deformation of the germ. Hence if the miniversal base of some germ of X has no smoothing component, then no such deformation of X (in particular no smoothing) has smooth general fibre.*

Lemma 2.3 (the cascade; [Wue26a, Lemma 2.4]). *Let $Q = R_0 + \cdots + R_r$ be a Minkowski decomposition of a unit-edge polygon into lattice summands, and consider Altmann's homogeneous deformation of $C(Q)$ attached to it, realized on the Cayley cone* $\mathrm{Cone}\big(\bigcup_i R_i \times \{f_i\}\big)$. *Its general fibre has exactly one singular point of type $C(R_i)$ for each two-dimensional non-smooth summand R_i, and no others. In particular the general fibre is* smooth *when every summand is a unit segment or a standard triangle.*

Proposition 2.4 (the singular locus of X; [Wue26a, Prop. 3.1]). *Let Δ be reflexive with unit edges and let $X \in |-K|$ be generic. Then X is a Calabi–Yau threefold with Gorenstein canonical singularities, and for each two-dimensional face F with $C(F)$* singular, *X meets the toric curve $V(\sigma_F)$ in exactly $\ell(F^\circ)$ points, all in the open orbit, at each of which an implicit-function argument on the chart $U_{\sigma_F} \cong C(F) \times \mathbb{C}^*$ identifies the germ of X with $C(F)$. Away from these points X is smooth.*

Throughout, Δ is the *fan* polytope, Batyrev's Δ^*, so $\mathbb{P}_\Delta$ is the toric fourfold of its face fan and our Δ° is his Δ, the Newton polytope of the anticanonical sections. We use the standard normalization $\Delta^\circ = \{u : \langle u, v\rangle \geq -1\ \forall v \in \Delta\}$, so that a facet of Δ corresponds to the vertex u of Δ° with $\langle u, \cdot\rangle = -1$ on the facet, and the dual edge of a two-dimensional face F is $F^\circ = \mathrm{conv}\{u_1, u_2\} \subset \Delta^\circ$ for the two facets through F. ([Wue26a] and the ancillary programs list facet data in the opposite normalization $\langle \cdot, \cdot\rangle = +1$; lattice lengths are unaffected.)

3. Homogeneous deformations attached to two-dimensional faces

Petracci [Pet21] constructs, from a *deformation datum* for a cone, homogeneous deformations of affine toric varieties ([Pet21, Def. 3.1, Thm. 3.5], after Mavlyutov [Mav04]) and of polarized projective toric pairs ([Pet21, Thm. 6.1]). Here N is a lattice, $M = \mathrm{Hom}(N, \mathbb{Z})$ its dual, and $\sigma \subseteq N \otimes \mathbb{R}$ a cone. A ∂-deformation datum for (N, σ), with σ strongly convex and full-dimensional, is a tuple $(Q, Q_0, \ldots, Q_r, w)$, $w \in M$, with (i) $Q \subseteq \sigma$ a non-empty rational polyhedron, (ii) $0 \notin Q$, (iii) $Q = Q_0 + \cdots + Q_r$, (iv) $Q_1, \ldots, Q_r$ lattice polyhedra, (v) $\min_Q w \geq -1$, and (vi) every vertex of $\sigma \cap \{\langle w, \cdot\rangle = -1\}$ contained in $\mathbb{R}^+ Q$. This is the boundary form of the definition, which is what [Pet21, Thm. 6.1] takes as input.

Lemma 3.1 (∂-deformation datum from a codimension-two face). *Let $\Delta \subset \mathbb{Z}^4$ be reflexive, F a two-dimensional face with a Minkowski decomposition $F = Q_0 + \cdots + Q_r$ into lattice polytopes, and let $\tau = \mathrm{Cone}(\Delta \times \{1\}) \subset \mathbb{Z}^5$, the cone over the polarized pair $(\mathbb{P}_\Delta, \partial\mathbb{P}_\Delta)$, where $\mathbb{P}_\Delta$ is the toric fourfold of the* face fan *of Δ as in §1. Let $u_1, u_2 \in \Delta^\circ$ be the vertices dual to the two facets of Δ containing F, so that the dual edge is $F^\circ = \mathrm{conv}\{u_1, u_2\}$, and suppose $\ell(F^\circ) \geq 2$. Then for every* interior *lattice point v^* of F°, the tuple*

$$\big(F \times \{1\},\ Q_0 \times \{1\},\ Q_1 \times \{0\}, \ldots, Q_r \times \{0\},\ w = (v^*, 0)\big)$$

is a ∂-deformation datum for $(\mathbb{Z}^5, \tau)$ in the sense of [Pet21, Def. 3.1].

Proof. Conditions (i)–(iv) are immediate ($Q = F \times \{1\}$ lies in τ at height one, and the summands may be translated so that $Q_1, \ldots, Q_r$ are lattice polytopes and Q_0 carries the height). For (v) and (vi), evaluate w on the rays of τ, which are $(V, 1)$ for the vertices V of Δ: $\langle w, (V, 1)\rangle = \langle v^*, V\rangle \geq -1$ because $v^* \in \Delta^\circ$, with equality if and only if V lies on the face of Δ dual to the face of Δ° containing v^* in its relative interior. Since v^* is an interior lattice point of the edge F°, that dual face is exactly F: equality holds precisely at the vertices of F. Hence $w \geq -1$ on the primitive generators of the rays of τ, with equality exactly for those in $\mathrm{Cone}(F \times \{1\})$; in particular $w \equiv -1$ on $Q = F \times \{1\}$, so $\min_Q w = -1$, which is (v). Moreover the vertices of the slice $\tau \cap \{w = -1\}$ are the points $(V, 1)$, $V \in \mathrm{vert}(F)$: a vertex of the slice lies on a ray of τ, whose w-value is an integer ≥ -1 and negative only if equal to -1. So (vi) holds with $\mathbb{R}^+Q = \operatorname{Cone}(F \times \{1\})$. $\square$

Remark 3.2. At the two endpoints of F° the slice grows to a full facet of Δ and condition (vi) would require a Minkowski decomposition of that three-dimensional facet; the interior points of the dual edge are exactly what makes the face-local datum possible. This is the source of the threshold $\ell(F^\circ) \geq 2$ in Theorem 1.1, and §5 shows the threshold is sharp.

By [Pet21, Thm. 6.1] the datum of Lemma 3.1 yields a flat deformation $(\mathcal{X}, \mathcal{D}) \to T$, T a neighbourhood of $0 \in \mathbb{A}^r$ (cf. [Pet21, Rem. 3.6]), of the pair $(\mathbb{P}_\Delta, \partial\mathbb{P}_\Delta)$, cut out by explicit trinomials in the Cox coordinates of a projective toric variety built from the extended cone $\tilde{\tau} = \operatorname{Cone}\big(\tau,\ Q_0 \times \{1\} - e_1 - \cdots - e_r,\ Q_1 + e_1, \ldots, Q_r + e_r\big) \subset \mathbb{Z}^{5+r}$. We call this the *face family* of $(F, \sum Q_i, v^*)$.

Lemma 3.3 (chart identification)**.** *Localized at the points of X over F, the face family induces on the germ $C(F)$ the Altmann–Cayley family of the decomposition $F = Q_0 + \cdots + Q_r$ (times a smooth trivial factor); in particular its general fibre near these points is the general fibre of Lemma 2.3.*

Proof. Write $N = \mathbb{Z}^4$ for the fan lattice of $\mathbb{P}_\Delta$, let $\sigma_F = \operatorname{Cone}(F) \subset N_\mathbb{R}$ be the cone of the face fan over F, and let N_F be the saturated sublattice spanned by σ_F, so that $N_F \cong \mathbb{Z}^3$, $M_F = \operatorname{Hom}(N_F, \mathbb{Z})$, and $C(F) = \operatorname{Spec} \mathbb{C}[\sigma_F^\vee \cap M_F]$ as in [Wue26a, §2]. (In Lemma 3.1 the same face appeared as $\operatorname{Cone}(F \times \{1\}) \subset \mathbb{Z}^5$, on the cone over the pair; on passing to the projective toric variety and its charts, $\mathbb{Z}^5$ is replaced by the fan lattice N, and the auxiliary lattice $\mathbb{Z}^5 \oplus \mathbb{Z}^r$ of the extended cone by $N \oplus \mathbb{Z}^r$.) A choice of complement, $N = N_F \oplus \mathbb{Z}\zeta$, gives the chart $U_{\sigma_F} \cong C(F) \times \mathbb{C}^*$ of Proposition 2.4, with $\mathbb{C}^*$-coordinate the character dual to ζ. The choice matters, and we fix ζ with

$$N = N_F \oplus \mathbb{Z}\zeta, \qquad \langle v^*, \zeta \rangle = 0.$$

Such ζ exists: v^* restricts to a surjection $N_F \to \mathbb{Z}$ (it takes the value -1 at each vertex of F), so any generator ζ_0 of N/N_F may be corrected to $\zeta = \zeta_0 - n$ with $n \in N_F$ chosen so that $\langle v^*, n \rangle = \langle v^*, \zeta_0 \rangle$.

Step 1: reduction to the affine germ. At the vertices V of F the functional v^* takes the value -1 (proof of Lemma 3.1), so $v^*|_{N_F} = -u_F$, the *negative* of the Gorenstein degree of σ_F (which is exactly the degree in which the deformations of an isolated Gorenstein toric threefold singularity live [Alt95], cf. [Pet21, Rem. 3.7]), and $(F, Q_0, \ldots, Q_r, -u_F)$ is a deformation datum for (N_F, σ_F).

We first normalize the translations of the summands. Choose a vertex $v = q_0 + \cdots + q_r$ of F, where q_i is the vertex of Q_i exposed by a functional exposing v, and replace

$$Q_0 \quad \text{by} \quad Q_0 + q_1 + \cdots + q_r, \qquad Q_i \quad \text{by} \quad Q_i - q_i \quad (i \geq 1).$$

These integral translations sum to zero and hence preserve the Minkowski sum. Since the differences of points of every Q_i lie in $\operatorname{span}_\mathbb{R}(F - F) = \ker u_F$, after this normalization

$$u_F(Q_0) = 1, \qquad u_F(Q_i) = 0 \quad (i \geq 1).$$

It does not change the family up to isomorphism. Indeed, putting $a_i = -q_i$ for $i \geq 1$, the lattice shear

$$(n, k; z_1, \ldots, z_r) \longmapsto \big(n + \textstyle\sum_i z_i a_i, k; z_1, \ldots, z_r\big)$$

fixes the original cone τ and the base characters e_i^*, and carries the old extended cone to the one formed from the translated summands. We henceforth suppress the primes.

Write $\tilde\sigma_F = \tilde\tau \cap (N_F \oplus \mathbb{Z}^r)_{\mathbb{R}}$ for the extended cone, where N_F is embedded in $\mathbb{Z}^5$ by the height section $n \mapsto (n, u_F(n))$; the displayed normalization is exactly what places the split summand generators in this graph lattice. Restrict the face family to the chart at the face $G \preceq \tilde\tau$ lying over σ_F (the cone of Step 2). To see that it is a face, use the functional

$$L = (v^*, 1, 0, \dots, 0).$$

On a ray $(V, 1)$ of τ it has value $\langle v^*, V\rangle + 1 \geq 0$, with equality exactly for $V \in F$. On the normalized split generators it vanishes because $v^*|_{N_F} = -u_F$, $u_F(Q_0) = 1$, and $u_F(Q_i) = 0$ for $i \geq 1$. Thus its zero cone is precisely G: each ray over a vertex $v = q_0 + \cdots + q_r$ of F is the sum of the corresponding split generators. On the chart's torus the j-th trinomial of the family reads

$$\chi^{e_j^*} - 1 \;=\; t_j\, \chi^{\tilde w}, \qquad \tilde w|_N = v^*$$

(divide the trinomial of Step 3 by the unit $\chi^{m_{j,0}}$; that the t_j-coefficient carries the character of the datum weight w is [Pet21, Thm. 3.5 and Rem. 3.7]). Every exponent here vanishes on ζ: the e_j^* because they annihilate N, and $\tilde w$ because $\langle v^*, \zeta\rangle = 0$ by the choice above. All r equations are therefore pulled back along the projection $(N_F \oplus \mathbb{Z}^r) \oplus \mathbb{Z}\zeta \to N_F \oplus \mathbb{Z}^r$, i.e. constant in the $\mathbb{C}^*$-direction: the restriction of the family $(\mathcal{X}, \mathcal{D}) \to \mathbb{A}^r$ to this chart is

$$\big(\text{Petracci's affine family of } (N_F, \sigma_F,\ F = \textstyle\sum Q_i)\big) \;\times\; \mathbb{C}^*,$$

as in the facet case [CHP24, Prop. 2.12] (there the relevant cone is maximal, so no complement need be chosen). It therefore suffices to identify the affine family, a deformation of $C(F)$ over $\mathbb{A}^r$.

Step 2: the Cayley cone. Each vertex of F has a unique expression $q_0 + q_1 + \cdots + q_r$ with $q_i \in \operatorname{vert}(Q_i)$: the normal cone of a vertex of a Minkowski sum is the intersection of the normal cones of its summand faces, and it is two-dimensional, forcing every summand face to be a vertex (no hypothesis on the summands is used, a point Proposition 4.7 exploits); correspondingly the rays of $\tilde\sigma_F$ over F are the vectors (q_0, h_0) and (q_j, h_j) $(j = 1, \dots, r)$, where in the auxiliary coordinates $(\text{height}, e_1, \dots, e_r) \in \mathbb{Z}^{1+r}$ one has $h_0 = (1; -1, -1, \dots, -1)$ and $h_j = (0; \mathbf{e}_j)$. The $(r{+}1) \times (r{+}1)$ matrix with rows $h_0, h_1, \dots, h_r$ is $\left(\begin{smallmatrix} 1 & -\mathbf{1}^\top \\ 0 & I_r \end{smallmatrix}\right)$, which is unimodular; the inverse automorphism of $\mathbb{Z}^{1+r}$ sends h_i to the standard basis vector f_i, and hence carries the face $G = \operatorname{Cone}\big(\bigcup_i Q_i \times \{h_i\}\big)$ onto the standard Cayley cone $\operatorname{Cone}\big(\bigcup_i Q_i \times \{f_i\}\big)$ of the decomposition (for $r = 1$ this is the matrix $\left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$).

Step 3: the family is Altmann's. Under the identification of Step 2, the projection $(e_1^*, \dots, e_r^*)$ that defines Petracci's family [Pet21, Thm. 3.5] becomes the projection $(f_1^* - f_0^*, \dots, f_r^* - f_0^*)$ of the Cayley cone: dual bases transform by the inverse transpose of the matrix of Step 2, which sends e_j^* to the functional taking the value -1 on h_0, 1 on h_j and 0 on the remaining rows, that is, to $f_j^* - f_0^*$. This projection is precisely Altmann's homogeneous one-parameter-per-summand deformation of $C(F)$ attached to $F = \sum_i Q_i$ [Alt95], whose realization on the Cayley cone and whose general fibre are recalled in Lemma 2.3. (Our base coordinates are $\chi^{u_j} - \chi^{u_0}$, the negatives of the ones used there, a harmless sign change of the parameters.) Concretely, Petracci's j-th trinomial restricts on the chart of G to $\chi^{m_{j,1}} - \chi^{m_{j,0}} = t_j$, where $m_{j,0}, m_{j,1} \in \widetilde{M}$ are the characters dual to the f_0- and f_j-heights. (For the type-(S) pentagon $F = T + s$, the Minkowski sum of a standard triangle T and a unit segment s, the case $r = 1$ is worked out explicitly in the ancillary files:[1] the five vertices of F split uniquely, and $m_{1,0} = (0, 0, 1, 0, 0, 0)$, $m_{1,1} = m_{1,0} + e_1^*$.)

[1] `paper2_check.py`.

Step 4. By [Wue26a, Lemma 2.4] the general fibre of this family is the general fibre of the Altmann family of the decomposition: one singular point of type $C(Q_i)$ for each two-dimensional non-smooth summand, and no other singular points. This proves the lemma. □

4. Smoothing over one face, and over several

Theorem 4.1 (restatement of Theorem 1.1)**.** *Let Δ be reflexive with unit edges and let F be a two-dimensional face of type* (S)*, with a Minkowski decomposition $F = Q_0 + \cdots + Q_r$ into unit segments and standard triangles, and suppose $\ell(F^\circ) \geq 2$. Then there is an explicit r-parameter deformation of the pair $(\mathbb{P}_\Delta, \partial\mathbb{P}_\Delta)$, given by trinomial equations in Cox coordinates, whose induced deformation of the generic anticanonical X has generic fibre smooth in a neighbourhood of the $\ell(F^\circ)$ points of type $C(F)$. In particular, if F is the* unique *non-smooth two-dimensional face of Δ, then X is smoothable.*

We first record that the hypersurface follows the ambient family.

Lemma 4.2 (the anticanonical family)**.** *Let $(\mathcal{X}, \mathcal{D}) \to (T, 0)$, $T \subseteq \mathbb{A}^r$ a neighbourhood of the origin, be the face family of Lemma 3.1, with central fibre $(\mathbb{P}_\Delta, \partial\mathbb{P}_\Delta)$. Then for every section $s_0 \in H^0(\mathbb{P}_\Delta, -K)$ there is, after shrinking T, a section $s \in H^0\big(\mathcal{X}, \mathcal{O}_\mathcal{X}(\mathcal{D})\big)$ restricting to s_0, and its zero locus $\mathcal{Y} \subset \mathcal{X}$ is flat over T with $\mathcal{Y}_0 = \{s_0 = 0\}$.*

Proof. By [Pet21, Thm. 6.1] the boundary deforms along the family, so $\mathcal{D} \subset \mathcal{X}$ is flat over T with central fibre $\partial\mathbb{P}_\Delta$; it is a relative effective Cartier divisor after shrinking T, since $\mathcal{D}_0$ is Cartier ($\mathbb{P}_\Delta$ being Gorenstein), so by flatness and Nakayama the ideal sheaf of $\mathcal{D}$ is invertible along $\mathcal{D}_0$, with local generators restricting to nonzerodivisors on nearby fibres.

The nearby $\mathcal{D}_t$ are again anticanonical. Gorenstein-ness is open in a proper flat family, so after shrinking T the sheaf $\omega_{\mathcal{X}/T}$ is a line bundle, and $\mathcal{M} = \omega_{\mathcal{X}/T}(\mathcal{D})$ has $\mathcal{M}_0 \cong \mathcal{O}$ with $h^1(\mathbb{P}_\Delta, \mathcal{O}) = 0$; by cohomology and base change the trivializing section extends, and by properness its zero locus, missing $\mathcal{X}_0$, misses the nearby fibres. Hence $\omega_{\mathcal{X}_t}(\mathcal{D}_t) \cong \mathcal{O}_{\mathcal{X}_t}$, and $\mathcal{L} = \mathcal{O}_\mathcal{X}(\mathcal{D})$ is a relative anticanonical bundle.

Since $-K_{\mathbb{P}_\Delta}$ is ample on the Gorenstein Fano toric fourfold $\mathbb{P}_\Delta$, toric Kodaira vanishing gives $h^i(\mathbb{P}_\Delta, -K) = 0$ for $i > 0$, hence the same on nearby fibres by semicontinuity; as $\chi(\mathcal{X}_t, \mathcal{L}_t)$ is constant in a flat family, so is $h^0(\mathcal{X}_t, \mathcal{L}_t)$, and Grauert's criterion makes $\pi_*\mathcal{L}$ locally free with formation commuting with base change. So s_0 extends to a section s of $\mathcal{L}$, and its zero locus $\mathcal{Y}$ is flat over T by the slicing criterion [Mat89, Thm. 22.6]: it suffices that each s_t be a nonzerodivisor on $\mathcal{X}_t$. After shrinking T the fibres $\mathcal{X}_t$ are reduced (geometric reducedness is open in a proper flat family), and s_t vanishes on no irreducible component: a component in its zero locus would give $\dim \mathcal{Y}_t = 4$, while the locus $\{t : \dim \mathcal{Y}_t \geq 4\}$ is closed ($\mathcal{Y} \to T$ is proper) and misses 0, since $\dim \mathcal{Y}_0 = 3$. Each $\mathcal{Y}_t$ is an anticanonical Cartier divisor in the Gorenstein $\mathcal{X}_t$, so adjunction gives $\omega_{\mathcal{Y}_t} \cong \mathcal{O}$, and $h^1(\mathcal{Y}_t, \mathcal{O}) = 0$ near $t = 0$ by semicontinuity: the fibres are Calabi–Yau. □

The step that actually smooths the *hypersurface*, as opposed to the ambient germ, is the following. It is where the genericity of s_0 is used.

Lemma 4.3 (the hypersurface follows the ambient smoothing)**.** *Let F be of type* (S) *with decomposition $F = Q_0 + \cdots + Q_r$ into unit segments and standard triangles and $\ell(F^\circ) \geq 2$, let $(\mathcal{X}, \mathcal{D}) \to T$ be the corresponding face family, and let $\mathcal{Y} \to T$ be the anticanonical family of Lemma 4.2 through a generic $s_0 \in H^0(\mathbb{P}_\Delta, -K)$. Let p be one of the $\ell(F^\circ)$ points of $X = \mathcal{Y}_0$ lying over F. Then there are a neighbourhood U of p in $\mathcal{X}$ and a non-empty Zariski-open $T^\circ \subseteq T$ such that $\mathcal{Y}_t \cap U$ is smooth for every $t \in T^\circ$.*

Proof. By Lemma 3.3 there are a neighbourhood U of p in $\mathcal{X}$ and an isomorphism over T

$$U \;\cong\; \mathcal{C} \times \mathbb{C}^*,$$

where $\mathcal{C} \to T$ is Altmann's Cayley family of the decomposition $F = \sum_i Q_i$ and the $\mathbb{C}^*$ factor is the direction of the toric curve $V(\sigma_F)$; the central fibre is $C(F) \times \mathbb{C}^*$, and p corresponds to (o, z_0) with o the cone point of $C(F)$. Write z for the coordinate on the $\mathbb{C}^*$ factor.

Because s_0 is generic, Proposition 2.4 gives $\ell(F^\circ)$ *distinct* zeros of s_0 on $V(\sigma_F) \cong \mathbb{P}^1$; a section of a degree-$\ell(F^\circ)$ line bundle on $\mathbb{P}^1$ with $\ell(F^\circ)$ distinct zeros has only simple ones, so (trivializing $\mathcal{L}$ near p and regarding s_0 as a holomorphic function) $\partial s_0/\partial z\,(p) \neq 0$. By Weierstrass division in the variable z (valid with parameters over the complex space $C(F)$), $s_0 = (\text{unit}) \cdot (z - \varphi_0)$ near p for a holomorphic $\varphi_0 : W_0 \to \mathbb{C}^*$ on a neighbourhood W_0 of o in $C(F)$, so $\{s_0 = 0\}$ is there the graph of φ_0; this is how Proposition 2.4 identifies the germ of X at p with $C(F)$.

Non-vanishing of $\partial s_t/\partial z$ is an open condition on $U \times T$, and s_t depends holomorphically on t (Lemma 4.2), so after shrinking U and T we have $\partial s_t/\partial z \neq 0$ throughout U for every t. Weierstrass division applies fibrewise: near each of its points, $\mathcal{Y}_t \cap U$ is the graph of a holomorphic function on an open subset of $\mathcal{C}_t$; in particular, locally on U,

$$\mathcal{Y}_t \cap U \;\cong\; \mathcal{C}_t \cap (\text{an open set}).$$

So $\mathcal{Y}_t \cap U$ is smooth as soon as $\mathcal{C}_t$ is. Every summand Q_i being a unit segment or a standard triangle, all summand cones $C(Q_i)$ are smooth, so by Lemma 2.3 the general fibre $\mathcal{C}_t$ is smooth; this holds on a non-empty Zariski-open $T^\circ \subseteq T$. ☐

Proof of Theorem 4.1 (assembly). Let $(\mathcal{X}, \mathcal{D}) \to T$ be the face family and $\mathcal{Y} \to T$ the anti-canonical family of Lemma 4.2 through a generic s_0, so $X = \mathcal{Y}_0$.

The points over F are smoothed. This is Lemma 4.3, applied at each of the $\ell(F^\circ)$ points of X over F: intersecting the finitely many non-empty Zariski-open subsets T° so obtained, $\mathcal{Y}_t$ is smooth near all of them for generic small $t \in T$.

The unique-face case. Suppose F is the unique non-smooth two-dimensional face of Δ. Then X is smooth away from the $\ell(F^\circ)$ points over F (Proposition 2.4), so $\mathcal{Y}_0$ is smooth outside the former F-points; smoothness is open on the total space of the proper flat family $\mathcal{Y} \to T$, so after shrinking, $\mathcal{Y}_t$ is smooth away from a neighbourhood of those points for all small t, and smooth near them for generic t by the previous step. Restricting $\mathcal{Y}$ to a generic line through $0 \in T$ gives a flat proper family over a disc with special fibre X and smooth Calabi–Yau general fibre (Lemma 4.2): X is smoothable. ☐

Remark 4.4 (first-order persistence at the other faces). None of the results below use this remark; we record it because it explains why the face families of distinct faces do not interfere at first order, which is the evidence behind Question 8.2. The face family does not disturb the other germs at first order. By [Pet21, Rem. 3.7] (stated there for the affine construction of [Pet21, Thm. 3.5]; the projective family carries the same parameter weights, as its trinomials do), the Kodaira–Spencer image of the family of Lemma 3.1 lies in the weight-w component of T^1, with $w = (v^*, 0)$. For a face $F' \neq F$, the space $T^1_{C(F')}$ is concentrated in the degree $-u_{F'}$, the character taking the value -1 on all rays $(V', 1)$ of $\sigma_{F'}$ [Alt95]; but F' has a vertex V' not on F, where $\langle w, (V', 1)\rangle = \langle v^*, V'\rangle \geq 0 \neq -1$ (it is an integer ≥ -1, and $= -1$ exactly on F by the proof of Lemma 3.1). So the induced first-order deformation of the germ over F' vanishes; likewise no positive integer multiple of w equals $-u_{F'}$, which by Rim's equivariant structure on the miniversal deformation [Rim80] forces the higher-order Taylor coefficients of an equivariant classifying map into vanishing weight spaces as well. Upgrading this to constancy of the germs requires identifying the induced deformation of the affine chart along $V(\sigma_{F'})$ with an equivariant product family, as in Lemma 3.3; this is routine for faces F' disjoint from F, where the chart of the extended cone over $\sigma_{F'}$ is untouched by the construction, but we do not pursue it here.

Theorem 4.5 (restatement of Theorem 1.2)**.** *Let Δ be reflexive with unit edges and suppose that every non-smooth two-dimensional face $F_1, \dots, F_s$ of Δ is of type* (S) *(a Minkowski sum of unit segments and standard triangles), with $\ell(F_i^\circ) \geq 2$. Then:*

(1) *for every singular point p of X there is an explicit deformation of X whose generic fibre is smooth in a neighbourhood of p; consequently the locus $D_p \subset (B, 0)$ of points of the semiuniversal deformation of X whose fibres remain singular near p is a proper closed analytic subgerm;*
(2) *if the base germ $(B, 0)$ of the semiuniversal deformation of X is irreducible (for instance smooth), then X is smoothable;*
(3) *if every F_i is a unit parallelogram, then X is smoothable unconditionally.*

Proof. (i) Let p lie over the face F_i and apply Theorem 4.1 to F_i: the resulting family $\mathcal{Y} \to T_i$ has generic fibre smooth near p. A semiuniversal deformation $\mathcal{S} \to (B, 0)$ of the compact complex space X exists by Grauert [Gra74]; choose a representative, proper over B. The relative singular locus $\Sigma = \mathrm{Sing}(\mathcal{S}/B)$ is closed, meets the central fibre in the finite set $\mathrm{Sing}\, X$, and so (fibre dimension being upper semicontinuous and the map proper) is finite over B after shrinking. Then Σ decomposes as a disjoint union $\bigsqcup_q \Sigma_q$ over the singular points q of X, and each $D_q := \mathrm{im}(\Sigma_q \to B)$ is a closed analytic subgerm (finite maps are proper). The family $\mathcal{Y} \to T_i$ is induced from $\mathcal{S}$ by a classifying map $\gamma_i : (T_i, 0) \to (B, 0)$; for generic t the fibre is smooth near p, so $\gamma_i(t) \notin D_p$: the subgerm D_p is proper.

(ii) After the shrinking above, a fibre $\mathcal{S}_b$ is smooth if and only if $b \notin D := \bigcup_q D_q$: away from the Σ_q the total space is smooth over B because the central fibre is (openness of smoothness). If $(B, 0)$ is irreducible, D, a finite union of proper closed analytic subgerms, is a proper closed analytic subgerm, so its complement meets every neighbourhood of 0. By the curve selection lemma for analytic germs there is a holomorphic arc $\varphi : (\mathbb{D}, 0) \to (B, 0)$ with $\varphi(\mathbb{D} \setminus 0) \not\subset D$ (normalize B and slice by $\dim B - 1$ generic hyperplanes through a preimage of 0: the preimage of D has smaller dimension, so the resulting curve germ is not contained in it); an irreducible arc not contained in the closed analytic set D meets it in a discrete set, so after shrinking the disc $\varphi(t) \notin D$ for all $t \neq 0$, and $\varphi^*\mathcal{S}$ is a smoothing of X.

(iii) If every F_i is a unit parallelogram, X is a nodal Calabi–Yau threefold and the MPCP resolution $\widehat{X} \to X$ is a projective *small* resolution (each square face has no interior or edge lattice points, so the maximal triangulation inserts only the diagonal wall). The $\ell(F_i^\circ) \geq 2$ nodes over one face lie on the single toric curve $V(\sigma_{F_i}) \cong \mathbb{P}^1$, and their exceptional curves in the MPCP resolution all represent the same class in $H_2(\widehat{X}; \mathbb{Q})$: in the Batyrev–Kreuzer presentation of this group by rational linear relations among the vertices of Δ, each maps to the diagonal relation of the face F_i, the sum of the endpoints of one diagonal minus the sum of the endpoints of the other, Batyrev–Kreuzer's ρ_θ [BK10] (see §7); choosing integers $a_{i,1}, \dots, a_{i,\ell_i}$ (writing $\ell_i = \ell(F_i^\circ)$), all non-zero, with $\sum_j a_{i,j} = 0$ (possible exactly because $\ell_i \geq 2$), and summing over the faces produces a relation $\sum a_k[\gamma_k] = 0$ among *all* exceptional classes with every coefficient non-zero. By the smoothing criterion for nodal Calabi–Yau threefolds with projective small resolution, X is smoothable. The criterion is Friedman's first-order result [Fri86], upgraded to genuine smoothings by the unobstructedness of nodal

Calabi–Yau deformations due to Kawamata and Tian (see [RT09]; the criterion is Namikawa's [Nam02], as used by [BK10]). □

Remark 4.6. The naive strategy for several faces (deform along the F_1-family, then repeat for F_2) does not make sense as it stands: the fibres of a face family are no longer toric hypersurfaces, so Lemma 3.1 cannot be applied to them. Nor do we know how to combine the data for several faces into a single deformation: Petracci's construction takes one datum, and the extended cones of different faces interfere. This is the same difficulty that leads Corti–Hacking–Petracci, one dimension down, to assemble their per-facet admissible Minkowski data not torically but through the semiuniversal deformation of a partial resolution with unobstructed deformations [CHP24]. Whether Theorem 4.5(ii) holds unconditionally (equivalently, whether the irreducibility hypothesis can be removed) is Question 8.2 below; note that by Gross [Gro97] the deformation space of a Calabi–Yau threefold with canonical singularities can be singular, so the hypothesis is not automatic.

4.1. **Arbitrary Minkowski decompositions: the cascade realized globally.** Theorem 1.1 invoked the type (S) hypothesis only at the last step, to conclude that the local general fibre is smooth. The deformation datum and its chart identification are insensitive to the summand types, so the same family realizes globally the *cascade* of [Wue26a, Lemma 2.4]: it moves the singular points over F to the germs of the two-dimensional non-smooth summands of a chosen Minkowski decomposition.

Proposition 4.7 (global cascade)**.** *Let Δ be reflexive with unit edges whose* unique *non-smooth two-dimensional face F has a Minkowski decomposition $F = Q_0 + \cdots + Q_r$ into lattice polytopes and $\ell(F^\circ) \geq 2$. Then there is an explicit r-parameter deformation of the generic anticanonical X, induced by a trinomial deformation of the ambient toric pair, such that on a generic line through the origin of the base, the fibre over every sufficiently small $t \neq 0$ is a Calabi–Yau threefold that has exactly $\ell(F^\circ)$ points of type $C(Q_i)$ for each two-dimensional non-smooth summand Q_i, and no other singular points. In particular, if F is deformable-but-non-smoothable (type* (D)*), take the construction for a maximal decomposition $F = R_0 + \cdots + R_s$. Then X deforms to a Calabi–Yau threefold in which each of the $\ell(F^\circ)$ points of type $C(F)$ is replaced by a rigid point $C(R_j)$ for each two-dimensional non-smooth summand R_j; this deformed threefold is again non-smoothable.*

Proof. Lemma 3.1 builds the deformation datum from any Minkowski decomposition into lattice polytopes with $\ell(F^\circ) \geq 2$: its proof uses only that $F = \sum_i Q_i$ and $v^* \in \operatorname{relint} F^\circ$, never the summand types. Lemma 3.3 is likewise independent of those types: a vertex of a Minkowski sum decomposes uniquely into vertices of the summands (its normal cone is the intersection of theirs), so that step, and the unimodular-height identification with the Altmann–Cayley cone, hold for an arbitrary decomposition. By the cascade lemma [Wue26a, Lemma 2.4] the general fibre of the Altmann–Cayley family has exactly one singular point of type $C(Q_i)$ for each two-dimensional non-smooth summand Q_i, and is smooth elsewhere. Lemma 4.2 carries a generic anticanonical section along the family, with Calabi–Yau fibres, and the graph description in the proof of Lemma 4.3 (which uses only $\partial s_t/\partial z \neq 0$, never the summand types) identifies $\mathcal{Y}_t$, near each former F-point, with an open subset of the Altmann fibre $\mathcal{C}_t$. Away from the former F-points the fibres are smooth, since X is (the face F being the unique non-smooth face) and smoothness is open on the total space of the proper flat family. So the singular points of $\mathcal{Y}_t$ are *at most* the summand germs, and it remains to check that they actually occur: that the singular points of $\mathcal{C}_t$ do not escape the neighbourhood over which the graph description holds. The Altmann–Cayley family is equivariant for the one-parameter subgroup through the lattice point $n_0 = (x, 1, \ldots, 1)$, x a vertex of F, which lies in the Cayley cone (it is the sum of one ray generator from each summand family); every base parameter is a difference of the height characters χ^{u_i}, each of weight $\langle u_i, n_0\rangle = 1$, so the subgroup carries the fibre over t to the fibre over λt and $\operatorname{Sing}\mathcal{C}_{\lambda t} = \lambda \cdot \operatorname{Sing}\mathcal{C}_t$. Because n_0 lies in the cone, every character of the dual cone has non-negative n_0-weight, so $\lim_{\lambda\to 0} \lambda \cdot p$ exists for every point p of the affine total space; applied to points of the relative singular locus, which is closed, the limits lie in $\operatorname{Sing} C(F) = \{o\}$. Hence along any fixed direction in the base the singular points of $\mathcal{C}_{\lambda t}$ converge to the cone point as $\lambda \to 0$, and on a generic line through the origin, for all sufficiently small $t \neq 0$ they lie inside the neighbourhood of o over which $\mathcal{Y}_t$ is a graph (a single sheet over a neighbourhood of o uniform in small t, by the Weierstrass division of the proof of Lemma 4.3 applied at the central point of the total space), which transports each of them, with its germ $C(Q_i)$, to a singular point of $\mathcal{Y}_t$. When F is of type (D), take the decomposition to be maximal. Every summand is then Minkowski-indecomposable, so every two-dimensional non-smooth summand is rigid. Thus each $C(F)$ point becomes a collection of $C(R_j)$ points; since each $C(R_j)$ has no smoothing component (Theorem 2.1), the deformed threefold admits no smoothing by Lemma 2.2. □

Remark 4.8. The smallest instance is the type-(D) pentagon $Q_B = T + s$ of [Wue26a] (T a non-standard determinant-3 lattice triangle, whose cone $C(T)$ is the $\frac{1}{3}(1,1,1)$ point, and s a unit segment; its maximal decomposition is $T + s$ with the single rigid summand T), realized with $\ell(F^\circ) = 3$ as the unique non-smooth face of the polytope Δ_B of [Wue26a]. The Calabi–Yau threefold X_B, whose three singular points are all deformable yet non-smoothable, then deforms, through a family induced by explicit ambient trinomials, to a Calabi–Yau threefold with three ordinary $\frac{1}{3}(1,1,1)$ points, itself non-smoothable. (That the conditions of Lemma 3.1 hold for this non-standard summand is a finite verification.[2]) Thus a single deformable-but-non-smoothable Calabi–Yau threefold lies in the closure of a family of quotient-singular ones: an explicit deformation between two non-smoothable compact Calabi–Yau threefolds, neither of them smoothable. It is an obstructed analogue of a conifold transition: the local deformation genuinely moves, but only trades each non-smoothable germ for its rigid core, never reaching a smooth fibre.

5. Sharpness: a one-nodal Calabi–Yau threefold with no smoothing

Theorem 5.1 (restatement of Theorem 1.3)**.** *Let Δ be a reflexive 4-polytope with unit edges. If the singular locus of its generic anticanonical hypersurface $X \subset \mathbb{P}_\Delta$ consists of a single ordinary double point, then X admits no smoothing.*

The polytope Δ_N of Theorem 1.3 provides an explicit example: it has 7 vertices and unit edges; its unique non-smooth two-dimensional face is the unit square $\square \times \{(1,0)\}$, of type (S)*, with dual edge of lattice length 1; the generic anticanonical X_N is a Calabi–Yau threefold with exactly one ordinary double point; and X_N admits no smoothing. The MPCP resolution $\widehat{X}_N$ has $(h^{1,1}, h^{2,1}) = (3, 103)$ and Euler number -200.*

Proof. That Δ_N is reflexive with unit edges, and that its 21 two-dimensional faces comprise 20 standard triangles and the square with $\ell(F^\circ) = 1$, are finite computations; the Hodge numbers are given by Batyrev's formulas [Bat94, §4], evaluated in exact arithmetic. The ten vertices of Δ_N°, in the normalization of §2 (so that $\Delta_N = \{v : \langle u, v\rangle \geq -1$ for every vertex u

[2] `cascade_check.py`.

of $\Delta_N^\circ\}$, with equality exactly on the facet dual to u), are

$$\begin{array}{lllll} (0,0,-1,-1), & (0,0,-1,0), & (1,0,-1,-1), & (5,-2,-1,-1), & (0,1,0,-1), \\ (0,-2,-1,-1), & (0,-2,-1,4), & (-2,0,1,-1), & (-2,0,1,4), & (-2,5,6,-1), \end{array}$$

with the dual edge F° of the square face spanned by the first two.[3] By Proposition 2.4, X_N has exactly one singular point, an ordinary double point.

It remains to prove the uniform assertion. Let $X \subset \mathbb{P}_\Delta$ be any hypersurface satisfying its hypotheses. Proposition 2.4 and the toric description of an ordinary double point identify its unique singularity with the cone over a unit square whose dual edge has lattice length one. The square has no interior lattice points, so an MPCP subdivision of its cone inserts only a diagonal wall: the induced crepant resolution $\pi : \widehat{X} \to X$ is *small*, with exceptional locus a single rational curve γ, and it is projective because the MPCP subdivision is projective [Bat94]. If A is ample on $\widehat{X}$ then $A \cdot \gamma > 0$, so $[\gamma] \neq 0$ in $H_2(\widehat{X};\mathbb{Q})$. By Friedman's theorem [Fri86] (extended to all odd-dimensional nodal Calabi–Yau varieties in [RT09]), a nodal Calabi–Yau threefold with a projective small resolution is smoothable only if the exceptional classes satisfy a relation $\sum a_i[\gamma_i] = 0$ in $H_2(\widehat{X};\mathbb{Q})$ with all $a_i \neq 0$; with a single node this forces $[\gamma] = 0$. Hence X admits no smoothing, and the explicit X_N is a special case. □

Remark 5.2. No *polynomial* deformation (moving X_N inside $\mathbb{P}_{\Delta_N}$) can smooth the node, since every Δ-regular member has the same singularity structure by Proposition 2.4; so Theorem 5.1 says the non-polynomial directions of [Mav04] fail as well. Contrast the generic one-nodal quintic threefold, where the node smooths polynomially: there the small resolutions are non-Kähler and $[\gamma]$ vanishes rationally. Δ_N manufactures the opposite situation inside the Kreuzer–Skarke list. Non-smoothable nodal Calabi–Yau threefolds as such are not new: the local-to-global failure for nodes is the classical content of [Fri86], and obstructed nodal configurations appear in Namikawa's study of the local moduli of Calabi–Yau threefolds [Nam02]. What Δ_N adds is a one-node realization produced, and shown non-smoothable, by the combinatorics of a single reflexive polytope.

Remark 5.3 (relation to the Batyrev–Kreuzer census). The nearest relative of Theorem 5.1 is the criterion of [BK10] itself, and we state the relationship precisely. Every two-dimensional face of Δ_N is a standard triangle or a unit parallelogram, with at least one parallelogram, so Δ_N belongs to the class of 198,849 polytopes surveyed in §7. The Batyrev–Kreuzer criterion recalled there is a necessary *and* sufficient condition, and it constrains exactly the faces with $\ell(F^\circ) = 1$: the diagonal relation of such a face must lie in the span of the diagonal relations of the *other* square faces. The unique square face of Δ_N has $\ell(F^\circ) = 1$ and there are no other square faces, so the span is zero while the diagonal relation is not; their criterion therefore also returns *non-smoothable*, and Δ_N is one of the $198{,}849 - 30{,}241 = 168{,}608$ polytopes their criterion does not show to be smoothable. Theorem 5.1 is proved directly from [Fri86], without the census, and what it contributes is not the mechanism but the *instance*: an explicit minimal one — one node, seven vertices, Hodge numbers computed — exhibited for the single purpose of showing that the threshold $\ell(F^\circ) \geq 2$ of Theorem 1.1 cannot be lowered. We have not found such an example in the literature: [BK10] tabulate the polytopes their criterion shows to be smoothable and do not discuss the complement.

[3]The ancillary programs list the same data in the opposite normalization $\langle\cdot,\cdot\rangle = +1$ of [Wue26a], that is, negated.

6. The $\ell(F^\circ) = 1$ germs: where Friedman's and Gross's criteria fall silent

Proposition 6.1 (restatement of Proposition 1.4). *Let Δ be reflexive with unit edges whose unique non-smooth two-dimensional face F has exactly one interior lattice point, is of type* (S)*, and has $\ell(F^\circ) = 1$. Then the singular locus of X is one point, the anticanonical cone over $S \in \{\mathbb{P}^1 \times \mathbb{P}^1,\ \mathrm{dP}_7,\ \mathrm{dP}_6\}$, and an MPCP resolution $\widehat{X} \to X$ can be chosen whose exceptional locus is the smooth irreducible surface S, contracted to the point. However, the contraction has relative Picard number $\rho(\widehat{X}/X) = \rho(S) \in \{2, 3, 4\}$, so it factors in the projective category and is* never *a primitive contraction, and X is not $\mathbb{Q}$-factorial at the point. In particular neither Gross's theorem on primitive type II contractions* [Gro97, Thm. 5.8] *nor his smoothing theorem for $\mathbb{Q}$-factorial contractions* [Gro97, Thm. 4.3] *applies, and the smoothability of X is open (Question 8.1).*

Proof. A unit-edge polygon with exactly one interior lattice point is, up to equivalence, one of five, and the corresponding cones are the anticanonical cones over the five smooth toric del Pezzo surfaces ([Wue26a, Rem. 2.6], which rests on [Gro97, §5.5]); two of them, $\mathbb{P}^2$ and $\mathbb{F}_1$, are rigid, and the remaining three are of type (S): the cones over $\mathbb{P}^1 \times \mathbb{P}^1$ (deg 8), dP_7 (deg 7) and dP_6 (deg 6). So F is one of these three. An MPCP subdivision is a maximal *projective* crepant subdivision and is not unique; we choose one whose restriction to $\mathrm{Cone}(F)$ is the star subdivision at the interior lattice point q of F: a pulling triangulation of the boundary that pulls first at q is regular [DLRS10, §4.3], hence projective [CLS11, Ch. 15], and pulling at q first forces every maximal cell over F to contain q. The star subdivision of $\mathrm{Cone}(F)$ at q is unimodular: after translating q to the origin, each triangle spanned by an edge of F and the origin has determinant ± 1, a finite check for each of the three polygons.[4] Hence this MPCP resolution is smooth over the point, with irreducible exceptional divisor the smooth toric del Pezzo surface S (the toric surface of the star of q) contracted to the point, and it is an isomorphism elsewhere; all other faces being smooth and all edges unit, $\widehat{X}$ is smooth everywhere.

Now the negative statements. Let k be the number of vertices of F, so $\rho(S) = k - 2 \in \{2, 3, 4\}$. For each vertex v' of F the toric divisor $V(v') \cap \widehat{X}$ of $\widehat{X}$ restricts to S as the toric boundary curve of S corresponding to the edge (q, v') of the star subdivision; these boundary divisors generate $\mathrm{Pic}(S)_{\mathbb{R}}$, so the intersection pairing of divisors of $\widehat{X}$ with curves contained in S has rank $\rho(S)$. Conversely every contracted curve lies in S, so its relative numerical class is in the image of $N_1(S)$ and the relative Picard number is at most $\rho(S)$. Thus the classes of the contracted curves span a face of dimension $\rho(S) \geq 2$ in $N_1(\widehat{X})$: the relative Picard number of $\widehat{X} \to X$ is $\rho(S) \geq 2$. Since $K_{\widehat{X}} \equiv 0$, the cone theorem says nothing about these rays directly; apply it instead to the klt pair $(\widehat{X}, \varepsilon S)$ for small $\varepsilon > 0$: adjunction gives $S|_S = K_S$, and $-K_S$ is ample on the del Pezzo surface S, so $(K_{\widehat{X}} + \varepsilon S) \cdot \gamma = \varepsilon\, K_S \cdot \gamma < 0$ for every contracted curve γ (all of which lie in S). The relative cone and contraction theorems for the pair then make $\overline{NE}(\widehat{X}/X)$ rational polyhedral with contractible extremal rays, and $\rho(S) \geq 2$ yields a factorization $\widehat{X} \to Z \to X$ with both steps non-trivial, so the contraction is not primitive; and since every projective crepant resolution of X is connected to this one by flops [Kaw08], which preserve the relative Picard number, no choice is primitive. (Concretely, for $S = \mathrm{dP}_7$ or dP_6: a (-1)-curve $\gamma \subset S$ has $N_{\gamma/\widehat{X}} = N_{\gamma/S} \oplus \omega_S|_\gamma = \mathcal{O}(-1) \oplus \mathcal{O}(-1)$ and can be contracted separately.) Finally, the local class group of the germ $C(F)$ is the cokernel of $M_F \to \mathbb{Z}^k$ (one coordinate per ray of σ_F), of rank $k - 3 \geq 1$, and these local classes are realized

[4] `paper2_check.py`.

by algebraic divisors on X: under the chart identification of the planting proposition, the closures of the intersections of X with the toric divisors of $\mathbb{P}_\Delta$ through the point restrict to the toric divisors of $C(F)$, which generate its class group. A class of infinite order in $\mathrm{Cl}(C(F))$ stays of infinite order in the analytic local class group at the point: divisor classes on the punctured germ are computed on the $\mathbb{C}^*$-bundle (total space of $\omega_S) \setminus S$ over the del Pezzo surface S, whose Picard group is $\mathrm{Pic}(S)/\mathbb{Z}K_S$ in the algebraic and the analytic category alike, the same group as $\mathrm{Cl}(C(F))$; analytically, this follows from the exponential sequence, using $H^1(S, \mathcal{O}_S) = H^2(S, \mathcal{O}_S) = 0$, and the Gysin sequence for the associated $\mathbb{C}^*$-bundle. Such a class is therefore represented by an algebraic divisor that is not $\mathbb{Q}$-Cartier at the point, so X is not $\mathbb{Q}$-factorial. (The germ computation alone would not suffice: a generic one-nodal quintic threefold is $\mathbb{Q}$-factorial although its node is analytically non-factorial.) Thus the hypotheses of both [Gro97, Thm. 5.8] (primitivity) and [Gro97, Thm. 4.3] ($\mathbb{Q}$-factoriality of X) fail. $\square$

Remark 6.2. Gross's theorem [Gro97, Thm. 5.8] smooths exactly these germs when they are realized by *primitive* type II contractions; such realizations exist on non-toric Calabi–Yau threefolds, where the restriction $\mathrm{Pic}(\widehat{X}) \to \mathrm{Pic}(S)$ can have rank one, and are constructed and smoothed for all three germs by Kapustka–Kapustka and Kapustka [KK09, Kap09]. The theorem fails exactly when the exceptional surface is $\mathbb{P}^2$ or $\mathbb{F}_1$, which are precisely the two classes with one interior point that are *rigid* in the trichotomy (Theorem 2.1): the germ classification and the global smoothing problem see the same dichotomy. Proposition 6.1 says the Batyrev hypersurfaces never provide the primitive realization: at $\ell(F^\circ) = 1$ the toric mechanism of §3 produces no face datum, Friedman's obstruction has no known analogue for these germs, and Gross's theorems do not apply: the smoothability of these X is genuinely open (Question 8.1).

7. Conifold configurations and the Batyrev–Kreuzer census

Suppose every non-smooth two-dimensional face of Δ is a unit parallelogram, so that X is a nodal Calabi–Yau threefold; this is the setting of Batyrev–Kreuzer [BK10]. They identify $H_2(\widehat{X}; \mathbb{Q})$ with the space of rational linear relations among the vertices of Δ, and show that the $\ell(F^\circ)$ exceptional curves over one square face F all represent a single class $[\gamma_F]$ there: up to the sign fixed by the choice of small resolution, the diagonal relation of F, the sum of the endpoints of one diagonal minus the sum of the endpoints of the other [BK10]. (The comparison carries a change of polarity: [BK10] place the parallelogram faces on the polar polytope, so their 1-dimensional face θ is our dual edge F°, their 2-face θ° is our F, and their relation ρ_θ is our diagonal relation of F. Our $\ell(F^\circ)$ is their $k_\theta = l(\theta) - 1$, the number of nodes lying over the face.) Friedman's relation restricted to one face, $\sum_{i=1}^{\ell} a_i[\gamma_F] = 0$ with all $a_i \neq 0$, is therefore satisfiable if and only if $\ell = \ell(F^\circ) \geq 2$; this reproduces the threshold of Theorem 1.1 from homology alone. A polytope in this class is thus smoothable whenever all its square faces have $\ell(F^\circ) \geq 2$ (Theorem 1.2(iii): sum the per-face relations and apply the nodal smoothing criterion to the projective small MPCP resolution), while faces with $\ell(F^\circ) = 1$ require a *cross-face rescue*: the class $[\gamma_F]$ must be a rational combination of the classes over other faces.

We tabulated this over the per-vertex-count database copy described in [Wue26a, §6.1], together with the separately reconstructed 36-vertex polytope. Assume, as in [Wue26a, Proposition 6.1], that the files contain without repetition exactly the classification members in their stated vertex range. Under this database hypothesis, for each reflexive 4-polytope we test every two-dimensional face for being a standard triangle or a unit parallelogram, and compute $\ell(F^\circ)$ as the lattice length of the edge of Δ° dual to it. Our enumeration finds exactly 198,849 reflexive 4-polytopes in the Batyrev–Kreuzer class: every two-dimensional face a standard triangle or a unit parallelogram, and at least one parallelogram, so that X is nodal with at least one node (polytopes all of whose faces are standard triangles give smooth X and are excluded). The total is in perfect agreement with the count of Batyrev–Kreuzer [BK10], an independent check on both computations. Of these 198,849, only 3,774 (1.90%) have *all* their square faces of dual length $\ell(F^\circ) \geq 2$, and are thus smoothable by the per-face argument alone; the remaining 195,075 each carry at least one $\ell(F^\circ) = 1$ square. The all-$\ell \geq 2$ fraction falls sharply with the vertex count: 39 of the 43 such polytopes with at most 7 vertices are all-$\ell \geq 2$, against 100 of the 148,959 with 14 or more vertices. Batyrev–Kreuzer show that 30,241 of the 198,849 are smoothable, via Namikawa's criterion. Their criterion constrains only the faces with $\ell(F^\circ) = 1$: in the form stated in [BK10], X is smoothable if and only if for every square face with $\ell(F^\circ) = 1$ the diagonal relation of that face lies in the span of the diagonal relations of the other square faces; faces with $\ell(F^\circ) \geq 2$ impose no condition, since their repeated class cancels against itself. A polytope whose squares all have $\ell(F^\circ) \geq 2$ therefore passes the criterion vacuously, and the 3,774 all-$\ell \geq 2$ polytopes all lie among the 30,241: exactly 3,774 of Batyrev–Kreuzer's 30,241 smoothings, one in eight, are explained by the per-face mechanism, and each of the remaining 26,467 involves an $\ell(F^\circ) = 1$ square whose exceptional class is rescued by a homology relation involving the classes over other faces. Understanding the combinatorial structure of these cross-face relations (the analogue, for the Calabi–Yau hypersurface, of the admissible Minkowski data of [CHP24]) is the central open problem left by this paper.

8. Open questions

Question 8.1. Is a Calabi–Yau threefold whose unique singular point has germ $C(F)$ with F of type (S), $i(F) \geq 1$, and $\ell(F^\circ) = 1$ ever smoothable? The nodal case $i(F) = 0$ is answered negatively by Theorem 5.1. For $i(F) = 1$ (the anticanonical cones over $\mathbb{P}^1 \times \mathbb{P}^1$, dP_7, dP_6) Proposition 6.1 shows the toric resolution is never a primitive contraction and X is not $\mathbb{Q}$-factorial, so Gross's theorems do not apply; for $i(F) \geq 2$ the crepant resolution contracts a reducible divisor, a maximal crepant subdivision inserting one exceptional prime divisor per interior lattice point of F. In neither case is a Friedman-type obstruction known, and the construction of §3 supplies no deformation datum.

Question 8.2. Is Theorem 1.2(ii) unconditional? Equivalently: can the face families of several type-(S) faces be realized simultaneously (a codimension-two analogue of the admissible Minkowski data of [CHP24]), or is the semiuniversal base of a Batyrev hypersurface with only type-(S) faces always irreducible? The fibres of a face family are not toric, so the construction cannot be iterated naively (Remark 4.6).

Question 8.3. Combined criterion: is X smoothable if and only if every non-smooth two-dimensional face is of type (S) and the exceptional classes over the $\ell(F^\circ) = 1$ faces admit a Friedman-type rescue? Together with [Wue26a] this would decide smoothability for all unit-edge Batyrev hypersurfaces, and the resulting criterion is fast enough to run over the full Kreuzer–Skarke list.

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